Multivariate Calculus Lecture Notes
Section 15.4 – Integration in Polar, Cylindrical, and Spherical Coordinates

I. Polar coordinates
   a. Some double integrals are much easier to evaluate in polar coordinates.
      i. This is because some regions are much easier to describe in polar coordinates.
      ii. Consider the following two regions.
         1. Graphically:

         2. In rectangular coordinates the regions are $\{(x, y) : x^2 + y^2 \leq 1\}$ and $\{(x, y) : 1 \leq x^2 + y^2 \leq 4, y \geq 0\}$.
         3. But in polar the regions are $\{(r, \theta) : 0 \leq r \leq 1\}$, $\{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$.
         4. What makes the regions simpler in polar is that the variables are bounded by constants.
   b. How do polar partitions work?
      i. In rectangular coordinates a differential region dA is a rectangle:

      ii. In polar, though, a differential region dA is a section of a sector of a circle:

      iii. What is the area of such a region?
         1. Remember that the area of a sector of a circle is $\frac{1}{2}r^2\theta$, where r is the radius of the circle and θ is the angle of the sector.
         2. Let r_k be the radius of the point in the center of the region.

         3. Then the outer radius is $r_k + \frac{\Delta r}{2}$, and the inner radius is $r_k - \frac{\Delta r}{2}$.
         4. So the area of the region is

$$\frac{1}{2}\left(r_k + \frac{\Delta r}{2}\right)^2 \Delta\theta - \frac{1}{2}\left(r_k - \frac{\Delta r}{2}\right)^2 \Delta\theta = \frac{\Delta\theta}{2}(2r_k\Delta r) = r_k\Delta r\Delta\theta.$$

         5. In the limit, then, this becomes $dA = rdrd\theta$.
      iv. This means that if you want to integrate over a region in polar coordinates you have to include $rdrd\theta$ in the integrand.

c. Examples

i. Let's finish the problem of finding the volume of the region bounded by the bottom half $z = 2 - \sqrt{x^2 + y^2}$ of an elliptic cone and the elliptic paraboloid $z = x^2 + y^2$.

1. We had brought the problem to the point of integrating

$$\iint_R \left[2 - \sqrt{x^2 + y^2} - (x^2 + y^2)\right] dA,$$

where R is the region given by $x^2 + y^2 \leq 1$.

2. With all the expressions involving $x^2 + y^2$ in there, perhaps a switch to polar coordinates would help.
   - The polar bounds are $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 1$.
   - The integrand is $2 - r - r^2$.
3. Thus we have

$$\int_0^{2\pi} \int_0^1 (2 - r - r^2) r \, dr \, d\theta = \left[\int_0^{2\pi} d\theta\right] \left[\int_0^1 (2r - r^2 - r^3) \, dr\right]$$
$$= \theta|_0^{2\pi} \left[r^2 - \frac{1}{3}r^3 - \frac{1}{4}r^4\right]_0^1 = 2\pi \left(1 - \frac{1}{3} - \frac{1}{4}\right) = (2\pi)\frac{5}{12} = \frac{5\pi}{6}.$$

ii. Evaluate $\iint_R (3x + 4y^2) \, dA$, where R is the region in the upper half-plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

1. Graphically:

2. This region is neither vertically nor horizontally simple.
   - It would take three integrals if we integrated first with respect to y.
   - But we can do it with one double integral in polar coordinates.
   - In polar, the region is described by $\{(r, \theta) : 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$.
3. Thus we solve $\int_0^{\pi} \int_1^3 (3x + 4y^2) r dr d\theta$. (Don't forget the extra r!)
   - Our integrand is in terms of x and y, so we need to convert to it to an expression in terms of r and θ.
   - This produces $\int_0^{\pi} \int_1^3 (3r\cos\theta + 4r^2 \sin^2\theta) r dr d\theta$.
   - We have

$$\int_0^{\pi} \int_1^3 (3r^2\cos\theta + 4r^3 \sin^2\theta) dr d\theta = \int_0^{\pi} \left[r^3 \cos\theta + r^4 \sin^2\theta\right]_1^2 d\theta$$
$$= \int_0^{\pi} \left(7\cos\theta + 15\sin^2\theta\right) d\theta = \int_0^{\pi} \left(7\cos\theta + (15)\frac{1 - \cos 2\theta}{2}\right) d\theta$$
$$= 7\sin\theta + \frac{15}{2}\left(\theta - \frac{1}{2}\sin 2\theta\right)\Big|_0^{\pi} = \frac{15\pi}{2}.$$

4. The following two trig identities (the first of which we used) can be useful:

$$\sin^2\theta = \frac{1-\cos 2\theta}{2} \quad \cos^2\theta = \frac{1+\cos 2\theta}{2}.$$

iii. Find the volume of a sphere of radius a.
1. We can find the volume of the top half and multiply by 2.
   - The top half of the sphere is given by the equation $z = \sqrt{a^2 - x^2 - y^2}$.
   - In polar this is $z = \sqrt{a^2 - r^2}$.
2. What is the region of integration?
   - Think of projecting the top half of the sphere onto the xy-plane.
   - This is the region of integration.
   - We get the disk given by $x^2 + y^2 \leq a^2$.
   - Thus the polar bounds are $0 \leq \theta \leq 2\pi$, $0 \leq r \leq a$.
3. Therefore,

$$V = 2\int_0^{2\pi}\int_0^a \sqrt{a^2-r^2}\, r\, dr\, d\theta = 2\left[\int_0^{2\pi} d\theta\right]\left[\int_0^a \sqrt{a^2-r^2}\, r\, dr\right]$$
$$= 4\pi\left[-\frac{1}{3}(a^2-r^2)^{3/2}\right]_0^a = (4\pi)\frac{1}{3}a^3 = \frac{4\pi a^3}{3}.$$

II. Cylindrical coordinates
a. Remember that a point P is represented by (r, θ, z), where
   i. (r, θ) is the projection of the point onto the xy-plane,
   ii. r is the distance from the origin to the projection,
   iii. θ is the angle between the projection and the positive x-axis
   iv. z is the distance to the point from the xy-plane.

b. Volume element in cylindrical coordinates
   i. It looks like this:

   ii. So in cylindrical coordinates, $dV = r\, dr\, d\theta\, dz$.
c. Examples
   i. A solid Q lies within the cylinder $x^2+y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. The density at any point is proportional to the distance from the axis of the cylinder. Find the mass of Q.
      1. Graphically:

2. Converting the cylinder to cylindrical coordinates yields $r^2 = 1$, which we can convert to $r = 1$.
3. Converting the paraboloid to cylindrical coordinates yields $z = 1 - r^2$.
4. Thus our region of integration can be described as $\{(r, \theta, z) : 1 - r^2 \leq z \leq 4, 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$.
5. Integrand
   - The density is proportional to the distance from the cylinder's axis.
   - Thus the integrand is Kr for some constant K.
6. We solve

$$\int_0^{2\pi}\int_0^1\int_{1-r^2}^4 Kr^2 dz\,dr\,d\theta = K\left[\int_0^{2\pi} d\theta\right]\left[\int_0^1\int_{1-r^2}^4 r^2 dz\,dr\right] = 2K\pi\int_0^1 zr^2\big|_{1-r^2}^4\,dr$$
$$= 2K\pi\int_0^1 (4r^2 - r^2 + r^4)dr = 2K\pi\int_0^1 (3r^2 + r^4)dr = 2K\pi\left[r^3 + \frac{1}{5}r^5\right]_0^1$$
$$= 2K\pi\left(1 + \frac{1}{5}\right) = \frac{12\pi}{5}K.$$

ii. Convert the following integral to cylindrical coordinates and evaluate it:

$$\int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2)\,dz\,dy\,dx.$$

1. Graphically:

2. Bounds
   - z: r to 2
   - r: 0 to 2
   - θ: 0 to 2π
3. We have

$$\int_0^{2\pi}\int_0^2\int_r^2 r^3\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^2 zr^3\Big|_r^2\,dr\,d\theta = \int_0^{2\pi}\int_0^2 (2r^3 - r^4)\,dr\,d\theta$$
$$= \left[\int_0^{2\pi} d\theta\right]\left[\frac{r^4}{2} - \frac{r^5}{5}\right]_0^2 = 2\pi\left(8 - \frac{32}{5}\right) = \frac{16\pi}{5}.$$

III. Spherical coordinates

a. A point P is represented by (ρ, ϕ, θ), where
   i. ρ is the distance between P and the origin,
   ii. ϕ is the angle between the positive z-axis and the line segment $\overline{OP}$, $0 \leq \phi \leq \pi$,
   iii. θ is the same as the angle in cylindrical coordinates.

b. Volume element in spherical coordinates

i. Graphically:

ii. What is dV in spherical coordinates?
   1. The volume element, for small $\Delta\rho$, $\Delta\phi$, and $\Delta\theta$, is approximately a rectangular prism (box).
   2. One side clearly has length $\Delta\rho$.
   3. The other two sides are circle sectors.
      - One sector has circle radius ρ and angle $\Delta\phi$, so its length is $\rho\Delta\phi$.
      - The other sector has circle radius r and angle $\Delta\theta$, so its length is $r\Delta\theta = \rho\sin\phi\Delta\theta$.
   4. Thus we must have $dV = \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$.

c. Examples

i. Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}}\,dV$, where B is the unit ball centered at the origin.
   1. In spherical coordinates, B is the set of points $\{(\rho,\phi,\theta) : 0 \le \rho \le 1, 0 \le \theta \le 2\pi, 0 \le \phi \le \pi\}$.
   2. And the integrand is just e^{ρ^3}.
   3. Thus we want

$$\begin{aligned}\int_0^{2\pi}\int_0^{\pi}\int_0^1 e^{\rho^3}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta &= \left[\int_0^{2\pi} d\theta\right]\left[\int_0^{\pi}\sin\phi\,d\phi\right]\left[\int_0^1 e^{\rho^3}\rho^2\,d\rho\right] \\ &= 2\pi\left(-\cos\phi\Big|_0^{\pi}\right)\left(\frac{1}{3}e^{\rho^3}\Big|_0^1\right) = 2\pi(1+1)\left(\frac{1}{3}(e-1)\right) \\ &= \frac{4\pi(e-1)}{3}.\end{aligned}$$

ii. Find the volume of the solid that lies above the graph of $z = \sqrt{x^2+y^2}$ and below the graph of $x^2+y^2+z^2 = z$.
   1. What is the graph of $x^2+y^2+z^2 = z$?
      - Complete the square to get $x^2+y^2+z^2-z+\frac{1}{4} = \frac{1}{4} \implies x^2+y^2+\left(z-\frac{1}{2}\right)^2 = \frac{1}{4}$.
      - Thus we have a sphere of radius $\frac{1}{2}$ centered at $\left(0,0,\frac{1}{2}\right)$.
   2. Since the graph of $z = \sqrt{x^2+y^2}$ is the top half of a cone the graph we have is the following:

   3. Bounds
      - θ: 0 to 2π

- ρ
  - The lower bound is 0.
  - The upper bound comes from the top half of the sphere.
  - Rewriting the sphere equation, we have $\rho^2 = \rho\cos\phi \implies \rho = 0$ or $\rho = \cos\phi$.
  - Thus our upper bound is $\rho = \cos\phi$.
- ϕ
  - The lower bound is 0.
  - The upper bound is the angle associated with the cone.
  - Rewriting the cone equation, we have $z = r \implies \rho\cos\phi = \rho\sin\phi \implies \rho = 0$ or $\cos\phi = \sin\phi$.
  - Thus our upper bound is $\phi = \dfrac{\pi}{4}$.

4. Thus we have

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^{\cos\phi} \rho^2\sin\phi\,d\rho\,d\phi\,d\theta = \left[\int_0^{2\pi} d\theta\right]\left[\frac{1}{3}\int_0^{\pi/4}\rho^3\Big|_0^{\cos\phi}\sin\phi\,d\phi\right] = \frac{2\pi}{3}\int_0^{\pi/4}\cos^3\phi\sin\phi\,d\phi$$
$$= -\frac{\pi}{6}\cos^4\phi\Big|_0^{\pi/4} = -\frac{\pi}{6}\left(\frac{1}{4} - 1\right) = \frac{\pi}{8}.$$

Homework: Section 15.4 #3, 9, 16, 18, 25, 30, 34, 37, 45, 46, 52.